Alex Example

alex@example.invalid | LinkedIn | Website

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Hiring Team
Example Systems

Application: Software Engineer, Data Infrastructure

Dear Example Systems Hiring Team,

Your Software Engineer, Data Infrastructure role interests me because it combines reliable data processing with performance work on systems used by other engineers. My recent work has focused on two related problems: making a batch ingestion pipeline recover safely after partial failures, and reducing the cost of repeated analytical queries. These projects are directly relevant to your requirements for fault-tolerant pipelines, SQL performance, and production ownership.

In a document ingestion project, I owned the Python workers and PostgreSQL persistence layer. A worker could commit a record and then lose its acknowledgement, causing a retry to create a duplicate. I used a stable source identifier with a database uniqueness constraint, and committed each record together with its processing checkpoint in one transaction. Workers acknowledged a task only after that transaction completed. This allowed at-least-once delivery while keeping repeated processing safe, without relying on a separate in-memory deduplication cache. I tested recovery by terminating workers before and after commits and replaying the same input. Across those failure cases, the pipeline resumed from persisted checkpoints and produced one stored record per source identifier.

On a separate analytics service, I was responsible for investigating slow PostgreSQL queries as the event history grew. I used EXPLAIN ANALYZE to separate time spent scanning rows from time spent sorting, then added a composite index aligned with the account filter and timestamp ordering. I also replaced offset pagination with a cursor based on timestamp and record ID, preserving a deterministic order when timestamps matched. I compared the original and revised queries against the same dataset and checked result equivalence, including tied timestamps and empty pages. At a fixed load, p95 response time fell from 820 ms to 190 ms.

I would welcome the opportunity to discuss these projects and how I could contribute to the reliability and performance of your data infrastructure.

Sincerely,
Alex Example